

Kenar Narayaka

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EDUCATION

University of Oxford, Christ Church

Oxford, England

MCompPhil in Computer Science & Philosophy – **Ranked 2nd** in Christ Church, **8th** across Oxford

2025 – 2029

Relevant Modules: Probability, Linear Algebra, Discrete Maths, Algorithms & Data Structures, Functional Programming (Haskell)

Activities: Indonesia Society President, Christ Church CompSoc Treasurer, Asia-Pacific Society Events Officer, Christ Church Basketball

Loughborough College

Loughborough, England

A-Level Maths, Further Maths, Computer Science - A*AA

2023 – 2025

WORK EXPERIENCE

Centorum

Jun. 2026 - Aug. 2026

Data Analyst | Polars, PyArrow, DuckDB

- Consolidated **150M+ procurement records** from fragmented sources using memory-bounded Polars/PyArrow batches and partitioned Parquet storage. Enabled supplier matching and duplicate checks without loading the complete dataset into memory.
- Matched suppliers and contracts in DuckDB; incremental updates achieved **4× throughput** versus old code on the same data and hardware.
- Deduplicated records with combined IDs, keeping source links for anomaly review; achieved **98.2% precision** on labelled duplicate pairs.

Hitachi Astemo

Jun. 2024 - Aug. 2024

Software Engineer Intern | PHP, Fat-Free Framework, PostgreSQL

- Reduced average leave, overtime and staffing request turnaround from **7 days to 5 hours in HR pilots** using role-based approvals and dashboards. Streamlined preparation for **600 weekly submissions**, cutting manual work from **10 to 3 minutes per form**.
- Migrated **80k historical HR records** into PostgreSQL using staged validation and duplicate checks; reconciled **99.8%** of records.
- Addressed slow dashboard loads with SQL indexes, joins and pagination; p95 page-load time fell from **2.1 s to 280 ms**.

CapitOx

Jan. 2026 - Apr. 2026

Quant Academy - Oxford-Man Institute

- Built an independent Monte Carlo put-pricing project, comparing five estimators under generalised-normal mixtures.
- Tuned importance sampling on separate pilots using shifted proposals; cut default ATM empirical variance $9.46\times$ versus IID at **20,000 samples**
- Evaluated **13,500 estimates** across three return laws, strikes and matched payoff budgets; checked RMSE and interval coverage against numerical integration, tracked tuning cost, and measured tuning variability using **five independent** pilot groups with **20 repeats** each.

OpenAI

Oct. 2026 - Present

Campus Lead

- Selected as Campus Lead to organise sponsored workshops, project sessions and student showcases using ChatGPT and Codex.

PROJECTS

3D Fractal Simulator | JavaScript, three.js, GLSL

- Created a browser-based GPU fractal explorer to develop intuition for 3D fractals, with live shape, colour, camera, and lighting controls.
- Accelerated 1080p rendering from **36 to 60 FPS** with **45% fewer distance checks** using adaptive ray steps and cone marching.
- Benchmarked **10 views per fractal: 16.7 ms median / 19.8 ms p95** frame times and **>0.98 SSIM** image similarity against reference renders. Enabled **4K export** and live exploration without rebuilding meshes; awarded **69/70** in the OCR Computer Science NEA.

Minecraft Beta Multiplayer Tunnelling Mod | Rust, Java, Tokio, Netty

- Simplified server-JAR setup to **one-button hosting**, using Rust/Tokio and Java/Netty to relay Minecraft TCP connections over QUIC.
- Used backpressure to bound queued data under slow receivers and half-close handling to preserve replies after one side stopped sending.
- Reserved ports atomically with signed, single-use leases; no duplicates in **10,000 competing attempts**. Passed **4,350 earlier simulated faults**; Windows tests sustained **100 Mb/s over eight streams**, **<20 ms p95 relay overhead** and no byte mismatches in a **two-hour soak**.

Neural Implicit Geometry: Thin-Feature Reconstruction | Python, PyTorch, CUDA

- Investigated fine-detail 3D reconstruction under a memory budget, comparing SIREN and Fourier-feature models at similar sizes.
- Replayed high-error samples, cutting thin-torus sampled symmetric Chamfer distance **32.5%** versus mixed near-surface sampling.
- Compared **six geometry settings** with **2.3k–33.7k parameters** and matched query/training budgets using surface error, volume overlap and mesh components. Gains across **three seeds** did not generalise: replay underperformed on **three fresh settings across 75 trials**.

ADDITIONAL INFO

Technical Skills: Python, Rust, Java, JavaScript, TypeScript, SQL, PHP, Haskell, Scala, GLSL

Interests: Linux (Arch), Basketball, Hiking, Poker, Philosophy, Pokemon

Spoken Languages: English, Indonesian